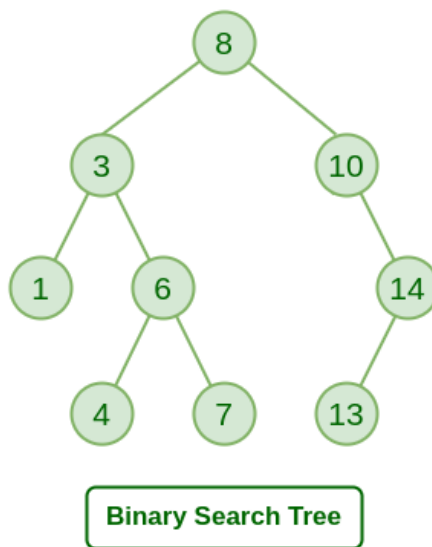


Binary Search Tree

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A **Binary Search Tree (or BST)** is a data structure used in computer science for organizing and storing data in a sorted manner. Each node in a **Binary Search Tree** has at most two children, a **left** child and a **right** child, with the **left** child containing values less than the parent node and the **right** child containing values greater than the parent node. This hierarchical structure allows for efficient **searching**, **insertion**, and **deletion** operations on the data stored in the tree.



Binary Search Tree

Introduction to Binary Search:

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- [Applications of BST](#)

Basic Operations on BST:

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- [Maximum in BST](#)
- [Floor in BST](#)

- [Ceil in BST](#)
- [Inorder Successor in BST](#)
- [Inorder Predecessor in BST](#)
- [Handling duplicates in BST](#)

Easy Standard Problems on BST:

- [Second largest in BST](#)
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- [BST keys in given Range](#)
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- [Binary Tree to BST](#)
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Medium Standard Problems on BST:

- [BST from Preorder](#)
- [Sorted Linked List to Balanced BST](#)
- [Transform a BST to greater sum tree](#)
- [BST to a Tree with sum of all smaller keys](#)
- [Construct BST from Level Order](#)
- [Check if an array can represent Level Order of BST](#)
- [Max Sum with No Two Adjacent in BST](#)
- [LCA in a BST](#)
- [Distance between two nodes of a BST](#)
- [k-th Smallest in BST](#)
- [Largest BST in a Binary Tree | Set 2](#)
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- [2 sum in BST](#)

- [Max between two nodes of BST](#)
- [Largest BST Subtree](#)
- [2 Sum in a Balanced BST](#)
- [Two nodes of a BST are swapped, correct it](#)
- [Leaf nodes from Preorder of a BST](#)

Hard Standard Problems on BST:

- [Construct all possible BSTs for keys 1 to N](#)
- [In-place Convert BST into a Min-Heap](#)
- [Check given array of size n can represent BST of n levels or not](#)
- [Merge two BSTs with limited extra space](#)
- [K'th Largest Element in BST when modification to BST is not allowed](#)
- [Check if given sorted sub-sequence exists in binary search tree](#)
- [Maximum Unique Element in every subarray of size K](#)
- [Count pairs from two BSTs whose sum is equal to a given value x](#)
- [Find if there is a triplet in a Balanced BST that adds to zero](#)
- [Replace every element with the least greater element on its right](#)
- [Leaf nodes from Preorder of a Binary Search Tree](#)
- [Minimum Possible value of \$|a_i + a_j - k|\$ for given array and k.](#)
- [Special two digit numbers in a Binary Search Tree](#)
- [Merge Two Balanced Binary Search Trees](#)

Some Quizzes:

- [‘Quizzes’ on Binary Search Tree](#)
- [‘Quizzes’ on Balanced Binary Search Trees](#)

Quick Links :

- [‘Practice Problems’ on Binary Search Tree](#)
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Introduction to Binary Search Tree

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Given a BST, the task is to insert a new node in this BST.Example: How to Insert a value in a Binary Search Tree:A new key is always inserted at the leaf by maintaining the property of the binary search...

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Searching in Binary Search Tree (BST)

Given a BST, the task is to search a node in this BST. For searching a value in BST, consider it as a sorted array. Now we can easily perform search operation in BST using Binary Search Algorithm. Input: Root of...

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Deletion in Binary Search Tree (BST)

Given a BST, the task is to delete a node in this BST, which can be broken down into 3 scenarios: Case 1. Delete a Leaf Node in BST Deletion in BST Case 2. Delete a Node with Single Child in BST Deleting a...

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Binary Search Tree (BST) Traversals – Inorder, Preorder, Post Order

Given a Binary Search Tree, The task is to print the elements in inorder, preorder, and postorder traversal of the Binary Search Tree. Input: A Binary Search Tree Output: Inorder Traversal: 10 20 30 100 150...

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Balance a Binary Search Tree

Given a BST (Binary Search Tree) that may be unbalanced, the task is to convert it into a balanced BST that has the minimum possible height. Examples: Input: Output: Explanation: The above unbalanced BST...

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